

SOLVED PRACTICE WORKBOOK

Rates-Motion-Time-and-Geometry - CSAT Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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BASIC MCQS / REMEDIATION

Diagnostic set - Questions 1-16

Q1. Practice variant 1: A finishes a job in 12 days and B in 18 days. Working together, about how many days do they take (nearest day)?

A. 7 B. 8 C. 6 D. 9

Q2. Practice variant 1: A pipe fills a tank in 10 hours and a leak empties it in 20 hours. How many hours together (nearest hour)?

A. 21 B. 20 C. 19 D. 22

Q3. Practice variant 1: A vehicle travels at 45 km/h for 4 hours. What distance does it cover?

A. 181 B. 179 C. 180 D. 182

Q4. Practice variant 1: A train 120 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 9 B. 7 C. 10 D. 8

Q5. Practice variant 1: A boat moves at 12 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 14 B. 15 C. 13 D. 16

Q6. Practice variant 1: What is the smaller angle between clock hands at 2:20?

A. 51 B. 50 C. 49 D. 52

Q7. Practice variant 1: If today is Monday, what weekday is it after 100 days?

A. Tuesday B. Thursday C. Wednesday D. Friday

Q8. Practice variant 1: What is the area in square metres of a rectangle 12 m by 8 m?

A. 97 B. 95 C. 98 D. 96

Q9. Practice variant 2: A finishes a job in 13 days and B in 19 days. Working together, about how many days do they take (nearest day)?

A. 8 B. 9 C. 7 D. 10

Q10. Practice variant 2: A pipe fills a tank in 11 hours and a leak empties it in 22 hours. How many hours together (nearest hour)?

A. 23 B. 22 C. 21 D. 24

Q11. Practice variant 2: A vehicle travels at 50 km/h for 4 hours. What distance does it cover?

A. 201 B. 199 C. 200 D. 202

Q12. Practice variant 2: A train 130 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 10 B. 8 C. 11 D. 9

Q13. Practice variant 2: A boat moves at 13 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 15 B. 16 C. 14 D. 17

Q14. Practice variant 2: What is the smaller angle between clock hands at 3:20?

A. 21 B. 20 C. 19 D. 22

Q15. Practice variant 2: If today is Monday, what weekday is it after 101 days?

A. Tuesday B. Wednesday C. Thursday D. Friday

Q16. Practice variant 2: What is the area in square metres of a rectangle 13 m by 9 m?

A. 118 B. 116 C. 119 D. 117

Graded set - Questions 17-32

Q17. Practice variant 3: A finishes a job in 14 days and B in 20 days. Working together, about how many days do they take (nearest day)?

A. 8 B. 9 C. 7 D. 10

Q18. Practice variant 3: A pipe fills a tank in 12 hours and a leak empties it in 24 hours. How many hours together (nearest hour)?

A. 25 B. 24 C. 23 D. 26

Q19. Practice variant 3: A vehicle travels at 55 km/h for 4 hours. What distance does it cover?

A. 221 B. 219 C. 220 D. 222

Q20. Practice variant 3: A train 140 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 10 B. 8 C. 11 D. 9

Q21. Practice variant 3: A boat moves at 14 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 16 B. 17 C. 15 D. 18

Q22. Practice variant 3: What is the smaller angle between clock hands at 4:20?

A. 11 B. 10 C. 9 D. 12

Q23. Practice variant 3: If today is Monday, what weekday is it after 102 days?

A. Tuesday B. Wednesday C. Friday D. Thursday

Q24. Practice variant 3: What is the area in square metres of a rectangle 14 m by 10 m?

A. 141 B. 139 C. 142 D. 140

Q25. Practice variant 4: A finishes a job in 15 days and B in 21 days. Working together, about how many days do they take (nearest day)?

A. 9 B. 10 C. 8 D. 11

Q26. Practice variant 4: A pipe fills a tank in 13 hours and a leak empties it in 26 hours. How many hours together (nearest hour)?

A. 27 B. 26 C. 25 D. 28

Q27. Practice variant 4: A vehicle travels at 60 km/h for 4 hours. What distance does it cover?

A. 241 B. 239 C. 240 D. 242

Q28. Practice variant 4: A train 150 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 11 B. 9 C. 12 D. 10

Q29. Practice variant 4: A boat moves at 15 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 17 B. 18 C. 16 D. 19

Q30. Practice variant 4: What is the smaller angle between clock hands at 5:20?

A. 41 B. 40 C. 39 D. 42

Q31. Practice variant 4: If today is Monday, what weekday is it after 103 days?

A. Tuesday B. Wednesday C. Saturday D. Thursday

Q32. Practice variant 4: What is the area in square metres of a rectangle 15 m by 11 m?

A. 166 B. 164 C. 167 D. 165

Remedial and timed set - Questions 33-48

Q33. Practice variant 5: A finishes a job in 16 days and B in 22 days. Working together, about how many days do they take (nearest day)?

A. 9 B. 10 C. 8 D. 11

Q34. Practice variant 5: A pipe fills a tank in 14 hours and a leak empties it in 28 hours. How many hours together (nearest hour)?

A. 29 B. 28 C. 27 D. 30

Q35. Practice variant 5: A vehicle travels at 65 km/h for 4 hours. What distance does it cover?

A. 261 B. 259 C. 260 D. 262

Q36. Practice variant 5: A train 160 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 12 B. 10 C. 13 D. 11

Q37. Practice variant 5: A boat moves at 16 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 18 B. 19 C. 17 D. 20

Q38. Practice variant 5: What is the smaller angle between clock hands at 6:20?

A. 71 B. 70 C. 69 D. 72

Q39. Practice variant 5: If today is Monday, what weekday is it after 104 days?

A. Tuesday B. Wednesday C. Sunday D. Thursday

Q40. Practice variant 5: What is the area in square metres of a rectangle 16 m by 12 m?

A. 193 B. 191 C. 194 D. 192

Q41. Practice variant 6: A finishes a job in 17 days and B in 23 days. Working together, about how many days do they take (nearest day)?

A. 10 B. 11 C. 9 D. 12

Q42. Practice variant 6: A pipe fills a tank in 15 hours and a leak empties it in 30 hours. How many hours together (nearest hour)?

A. 31 B. 30 C. 29 D. 32

Q43. Practice variant 6: A vehicle travels at 70 km/h for 4 hours. What distance does it cover?

A. 281 B. 279 C. 280 D. 282

Q44. Practice variant 6: A train 170 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 12 B. 10 C. 13 D. 11

Q45. Practice variant 6: A boat moves at 17 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 19 B. 20 C. 18 D. 21

Q46. Practice variant 6: What is the smaller angle between clock hands at 7:20?

A. 101 B. 100 C. 99 D. 102

Q47. Practice variant 6: If today is Monday, what weekday is it after 105 days?

A. Tuesday B. Wednesday C. Monday D. Thursday

Q48. Practice variant 6: What is the area in square metres of a rectangle 17 m by 13 m?

A. 222 B. 220 C. 223 D. 221

PYQS AND ANSWER PRACTICE

Complete step-by-step solutions

Q1. Practice variant 1: A finishes a job in 12 days and B in 18 days. Working together, about how many days do they take (nearest day)?

A. 7 B. 8 C. 6 D. 9

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q2. Practice variant 1: A pipe fills a tank in 10 hours and a leak empties it in 20 hours. How many hours together (nearest hour)?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Net rate equals fill rate minus leak rate.

Q3. Practice variant 1: A vehicle travels at 45 km/h for 4 hours. What distance does it cover?

A. 181 B. 179 C. 180 D. 182

Correct answer: C. Distance = speed x time after matching units.

Q4. Practice variant 1: A train 120 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 9 B. 7 C. 10 D. 8

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q5. Practice variant 1: A boat moves at 12 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 14 B. 15 C. 13 D. 16

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q6. Practice variant 1: What is the smaller angle between clock hands at 2:20?

A. 51 B. 50 C. 49 D. 52

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q7. Practice variant 1: If today is Monday, what weekday is it after 100 days?

A. Tuesday B. Thursday C. Wednesday D. Friday

Correct answer: C. Weekdays repeat modulo seven.

Q8. Practice variant 1: What is the area in square metres of a rectangle 12 m by 8 m?

A. 97 B. 95 C. 98 D. 96

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

Q9. Practice variant 2: A finishes a job in 13 days and B in 19 days. Working together, about how many days do they take (nearest day)?

A. 8 B. 9 C. 7 D. 10

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q10. Practice variant 2: A pipe fills a tank in 11 hours and a leak empties it in 22 hours. How many hours together (nearest hour)?

A. 23 B. 22 C. 21 D. 24

Correct answer: B. Net rate equals fill rate minus leak rate.

Q11. Practice variant 2: A vehicle travels at 50 km/h for 4 hours. What distance does it cover?

A. 201 B. 199 C. 200 D. 202

Correct answer: C. Distance = speed x time after matching units.

Q12. Practice variant 2: A train 130 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 10 B. 8 C. 11 D. 9

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q13. Practice variant 2: A boat moves at 13 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 15 B. 16 C. 14 D. 17

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q14. Practice variant 2: What is the smaller angle between clock hands at 3:20?

A. 21 B. 20 C. 19 D. 22

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q15. Practice variant 2: If today is Monday, what weekday is it after 101 days?

A. Tuesday B. Wednesday C. Thursday D. Friday

Correct answer: C. Weekdays repeat modulo seven.

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A. 118 B. 116 C. 119 D. 117

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A. 25 B. 24 C. 23 D. 26

Correct answer: B. Net rate equals fill rate minus leak rate.

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A. 221 B. 219 C. 220 D. 222

Correct answer: C. Distance = speed x time after matching units.

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A. 10 B. 8 C. 11 D. 9

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A. 16 B. 17 C. 15 D. 18

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q22. Practice variant 3: What is the smaller angle between clock hands at 4:20?

A. 11 B. 10 C. 9 D. 12

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q23. Practice variant 3: If today is Monday, what weekday is it after 102 days?

A. Tuesday B. Wednesday C. Friday D. Thursday

Correct answer: C. Weekdays repeat modulo seven.

Q24. Practice variant 3: What is the area in square metres of a rectangle 14 m by 10 m?

A. 141 B. 139 C. 142 D. 140

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

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A. 9 B. 10 C. 8 D. 11

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q26. Practice variant 4: A pipe fills a tank in 13 hours and a leak empties it in 26 hours. How many hours together (nearest hour)?

A. 27 B. 26 C. 25 D. 28

Correct answer: B. Net rate equals fill rate minus leak rate.

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A. 241 B. 239 C. 240 D. 242

Correct answer: C. Distance = speed x time after matching units.

Q28. Practice variant 4: A train 150 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 11 B. 9 C. 12 D. 10

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q29. Practice variant 4: A boat moves at 15 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 17 B. 18 C. 16 D. 19

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Q30. Practice variant 4: What is the smaller angle between clock hands at 5:20?

A. 41 B. 40 C. 39 D. 42

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

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A. Tuesday B. Wednesday C. Saturday D. Thursday

Correct answer: C. Weekdays repeat modulo seven.

Q32. Practice variant 4: What is the area in square metres of a rectangle 15 m by 11 m?

A. 166 B. 164 C. 167 D. 165

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

Q33. Practice variant 5: A finishes a job in 16 days and B in 22 days. Working together, about how many days do they take (nearest day)?

A. 9 B. 10 C. 8 D. 11

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q34. Practice variant 5: A pipe fills a tank in 14 hours and a leak empties it in 28 hours. How many hours together (nearest hour)?

A. 29 B. 28 C. 27 D. 30

Correct answer: B. Net rate equals fill rate minus leak rate.

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A. 261 B. 259 C. 260 D. 262

Correct answer: C. Distance = speed x time after matching units.

Q36. Practice variant 5: A train 160 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 12 B. 10 C. 13 D. 11

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A. 18 B. 19 C. 17 D. 20

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Q38. Practice variant 5: What is the smaller angle between clock hands at 6:20?

A. 71 B. 70 C. 69 D. 72

Correct answer: B. Hour hand angle is $30h + 0.5m$; minute hand angle is $6m$.

Q39. Practice variant 5: If today is Monday, what weekday is it after 104 days?

A. Tuesday B. Wednesday C. Sunday D. Thursday

Correct answer: C. Weekdays repeat modulo seven.

Q40. Practice variant 5: What is the area in square metres of a rectangle 16 m by 12 m?

A. 193 B. 191 C. 194 D. 192

Correct answer: D. Rectangle area = length x breadth; the unit is squared.

Q41. Practice variant 6: A finishes a job in 17 days and B in 23 days. Working together, about how many days do they take (nearest day)?

A. 10 B. 11 C. 9 D. 12

Correct answer: A. Add work rates $1/d_1 + 1/d_2$, then invert.

Q42. Practice variant 6: A pipe fills a tank in 15 hours and a leak empties it in 30 hours. How many hours together (nearest hour)?

A. 31 B. 30 C. 29 D. 32

Correct answer: B. Net rate equals fill rate minus leak rate.

Q43. Practice variant 6: A vehicle travels at 70 km/h for 4 hours. What distance does it cover?

A. 281 B. 279 C. 280 D. 282

Correct answer: C. Distance = speed x time after matching units.

Q44. Practice variant 6: A train 170 m long crosses a pole at 54 km/h. How many seconds does it take?

A. 12 B. 10 C. 13 D. 11

Correct answer: D. Convert km/h to m/s, then time = train length/speed.

Q45. Practice variant 6: A boat moves at 17 km/h in still water and the stream is 2 km/h. What is its downstream speed?

A. 19 B. 20 C. 18 D. 21

Correct answer: A. Downstream speed equals still-water speed plus stream speed.

Q46. Practice variant 6: What is the smaller angle between clock hands at 7:20?

A. 101 B. 100 C. 99 D. 102

Correct answer: B. Hour hand angle is $30h+0.5m$; minute hand angle is $6m$.**Q47. Practice variant 6: If today is Monday, what weekday is it after 105 days?**

A. Tuesday B. Wednesday C. Monday D. Thursday

Correct answer: C. Weekdays repeat modulo seven.**Q48. Practice variant 6: What is the area in square metres of a rectangle 17 m by 13 m?**

A. 222 B. 220 C. 223 D. 221

Correct answer: D. Rectangle area = length x breadth; the unit is squared.**Verified PYQ routing ledger**

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2024	5	Cube partition cuts	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	7	Work-rate scaling	C (supplied)	Add rates and invert the net rate.
2024	8	Alternating-worker schedule	C (supplied)	Add rates and invert the net rate.
2024	30	Metric-unit conversion	D (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2024	37	Clock-hand coincidence	A (supplied)	Use relative angular speed.
2024	38	Calendar repetition	C (supplied)	Reduce the day shift modulo seven.
2024	40	Clock-hand angle	C (supplied)	Use relative angular speed.
2025	9	Run-rate time relation	D (supplied)	Draw a labelled generation graph.
2025	15	Painted-cube partition	C (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	37	Circular-track encounters	B (supplied)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2025	39	Overtaking relative speed	C (supplied)	Use relative speed after unit conversion.
2025	45	Pipes-and-cistern rate	D (supplied)	Assign positive fill and negative empty rates.
2025	59	Fractional work rates	B (supplied)	Add rates and invert the net rate.
2026	13	Net displacement with jumps	D (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	20	Relative-speed train crossing	C (provisional)	Use relative speed after unit conversion.
2026	24	Cube-cutting sequence count	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	37	Circular-track relative motion	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	41	Sound-distance relative rate	C (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.

Year	Q	Neutral verified demand	Locally checked Set-A key/status	Solution architecture
2026	50	Multi-segment speed comparison	D (provisional)	Use relative speed after unit conversion.
2026	68	Rectangular-tile area count	A (provisional)	Classify the dominant mechanism, represent the givens, solve minimally and verify independently.
2026	69	Train-accident relative rate	A (provisional)	Use the total crossing length.

Final error-log checklist

- Was the family classified correctly?
- Were units, domains, quantifiers and constraints copied correctly?
- Was the first-principles relation written before a shortcut?
- Was the selected option independently recomputed or supported by the passage?
- Was the mistake logged and followed by a fresh remedial item?